EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 08

Basic Maths and Calculus

Graph Shapes and Physics Applications

By - Excellence Programme

11 September 2026

Topics to be covered

01 Spring-force graph and perpendicular lines

02 Modulus graphs and transformations

03 Rectangular hyperbola and inverse relations

04 Parabola, square-root graph and transformations

05 Graphs used in mechanics, electricity and thermal physics

06 Circle and S.H.M. homework with guided answers

Rough work / extra work

Equation of straight line

$$y = mx + c$$

m = slope of line

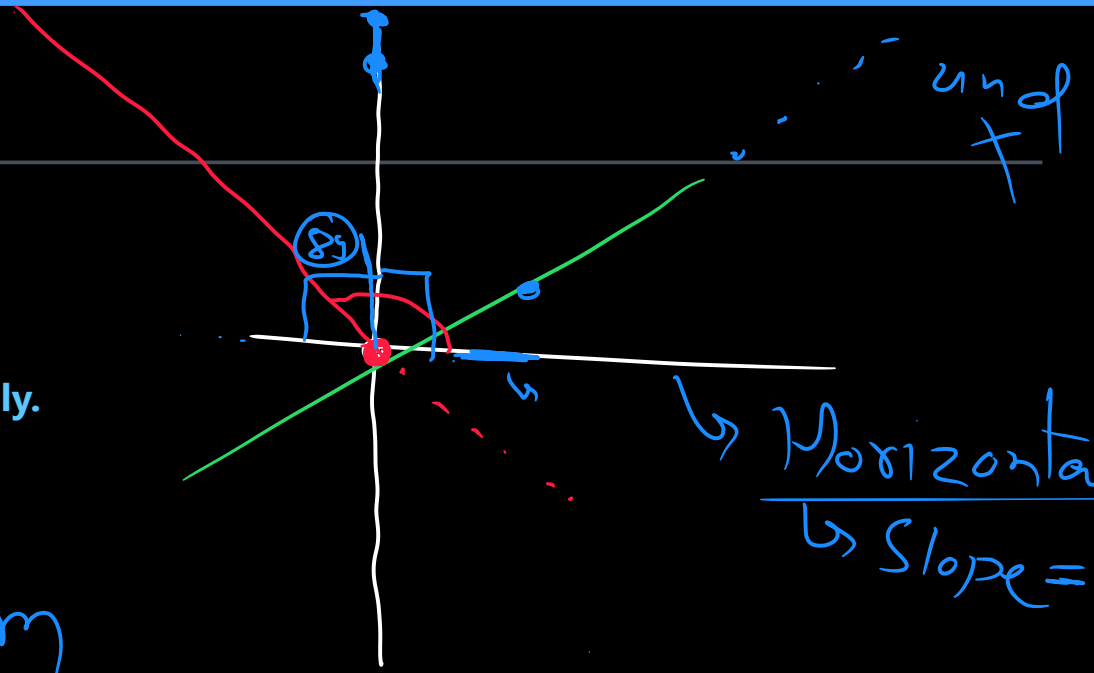
c = Intersect = Distance from origin on x-axis

Only the graph tools we need

rising line: positive slope **falling line: negative slope**

horizontal line: slope 0 **vertical line: slope undefined**

To draw any graph: make a small value table, mark points, join smoothly.



We measure from x-axis ⇒ so slope of x-axis always 0

Slope $= m = \tan\theta$

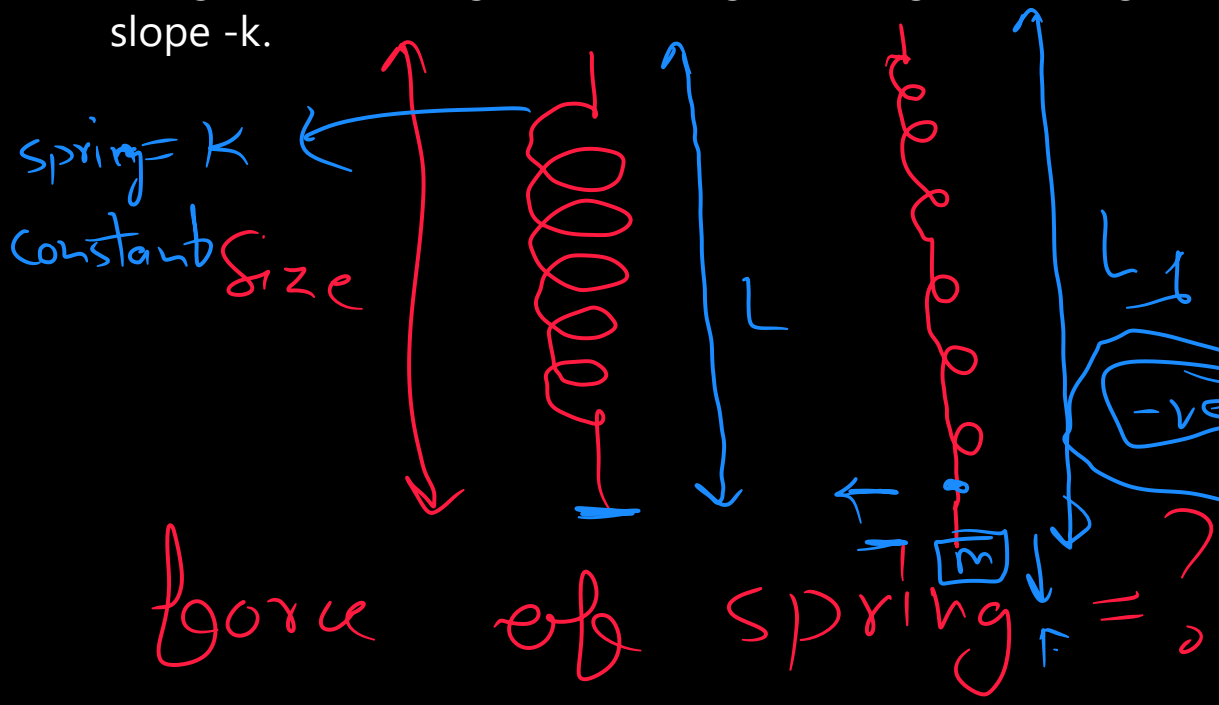
$= \dfrac{\tan 90}{\to \infty} = \dfrac{\sin 90}{\cos 90} = \dfrac{1}{0} = \infty / \text{undf}$ ⇒

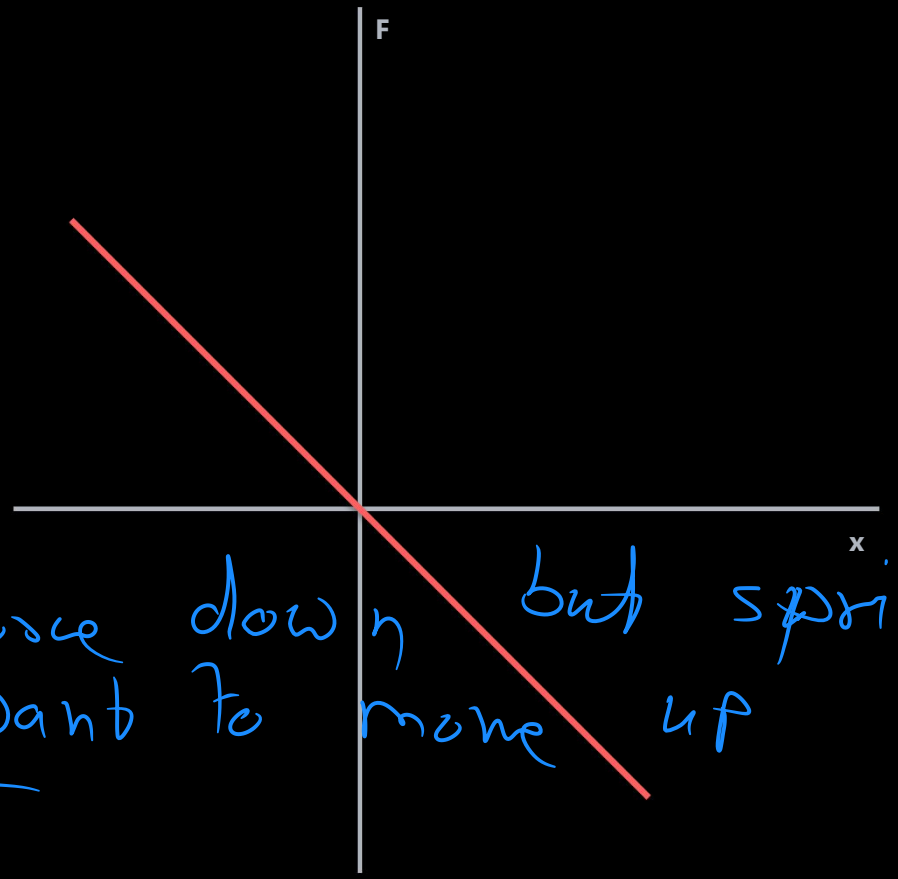
Spring force: F = -kx

Hooke's law says the spring force is opposite to displacement.

F = -kx

The graph is a straight line through the origin with negative slope -k.





Force down but spring Want to move up

$F = -(L_1 - L)k$

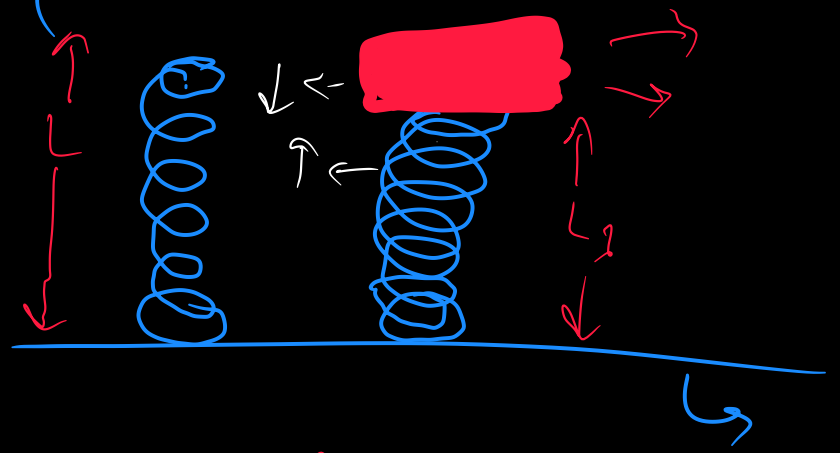
We add negtive coz x is Expanding the direction of -x

Theory of the spring-force graph

1. x is displacement from the equilibrium position. → is negative
2. k is the spring constant. A larger k means a stiffer spring.
3. The minus sign means force acts opposite to displacement.

x>0 gives F<0 and x<0 gives F>0

Slope of the F-x graph = -k. Its magnitude gives the spring constant.



In current situation force = Down

but If remove mass force

will be = UP

Perpendicular straight lines

Two perpendicular lines meet at 90 degrees.

m1 x m2 = -1

VIP

If m1 = tan(theta), then m2 = tan(theta+90 degrees) = -1/tan(theta)

This rule is used only when both slopes are defined.

$M_A = \tan\theta$

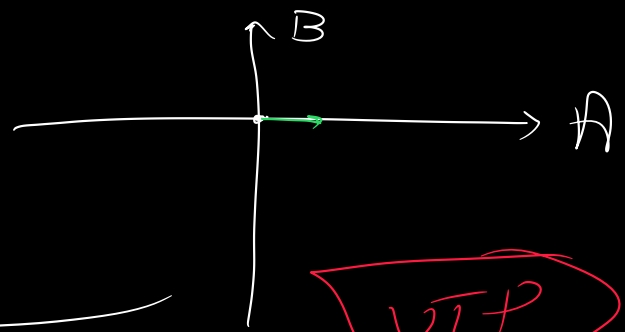
$M_B = \tan(90+\theta)$

$= -\cot\theta$

$M_A \times M_B = \tan\theta \times -\cot\theta$

$= \tan\theta \times \frac{-1}{\tan\theta}$

$= -1$



VIP

$\Leftarrow M_A = \tan 0 \Rightarrow$ If This θ Then

$M_B = \tan 90 \Rightarrow$ It will be

$\Rightarrow \tan(90+\theta)$

$\Rightarrow \tan(90+\theta) =$

$\Rightarrow \frac{\sin(90+\theta)}{\cos(90+\theta)} = \frac{\cos\theta}{-\sin\theta}$

$= -\cot\theta$

Theory of perpendicular slopes

Rotating a line by 90 degrees changes its direction completely.

m1=tan(theta)

m2=tan(theta+90 degrees)=-cot(theta)=-1/tan(theta)

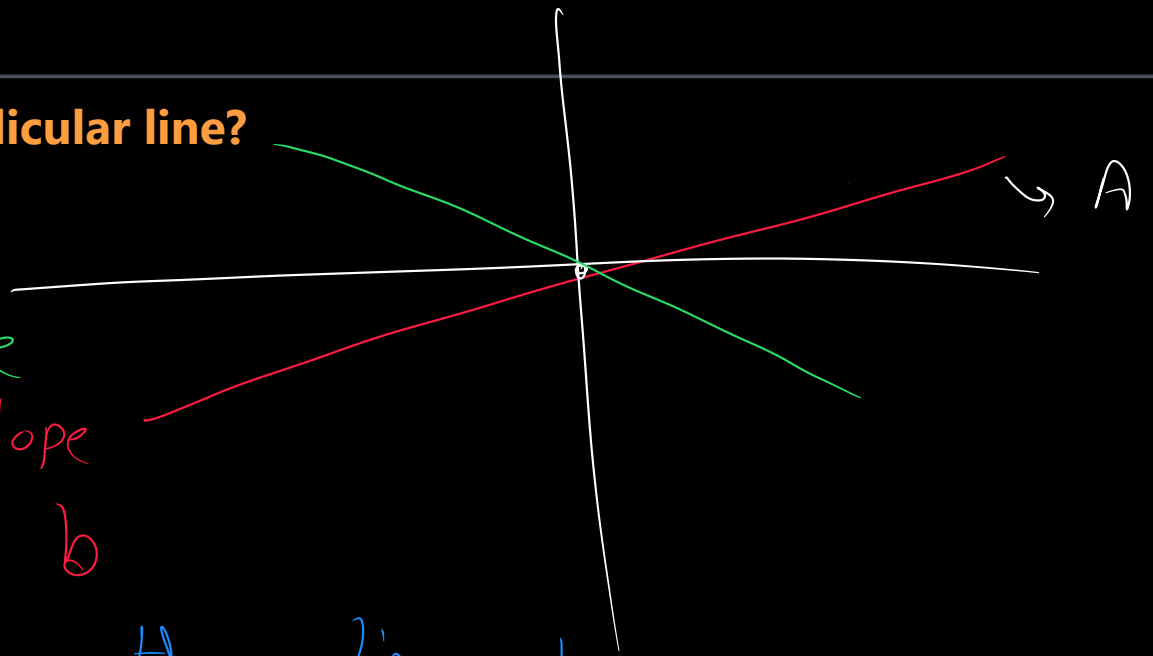
Therefore m1 x m2 = tan(theta) x [-1/tan(theta)] = -1

$$\Rightarrow \tan\theta \times \frac{-1}{\tan\theta} = -1$$

A horizontal line is perpendicular to a vertical line; the product rule cannot be used because vertical slope is undefined.

Perpendicular-line check

A line has slope 2. What is the slope of a perpendicular line?



Angle between these 2 line unknown but I know slope of line A is 2 & line b is $\frac{-1}{2}$ ⇒ Can we confirm these line ⊥ to each other?

$$m_A \times m_B = 2 \times \frac{-1}{2} = -1$$

So line ⊥ to each other

Simple solution

Step 1: Use m1 x m2 = -1

Step 2: 2 x m2 = -1

Step 3: m2 = -1/2

Perpendicular slope = -1/2

Rough work / extra work

If 2 lines ⊥ to each other & slope of one line is 3, what is slope of second line ? ⇒

We know $m_1 \times m_2 = -1$

& $m_1 = 3$

So $3 \times m_2 = -1$

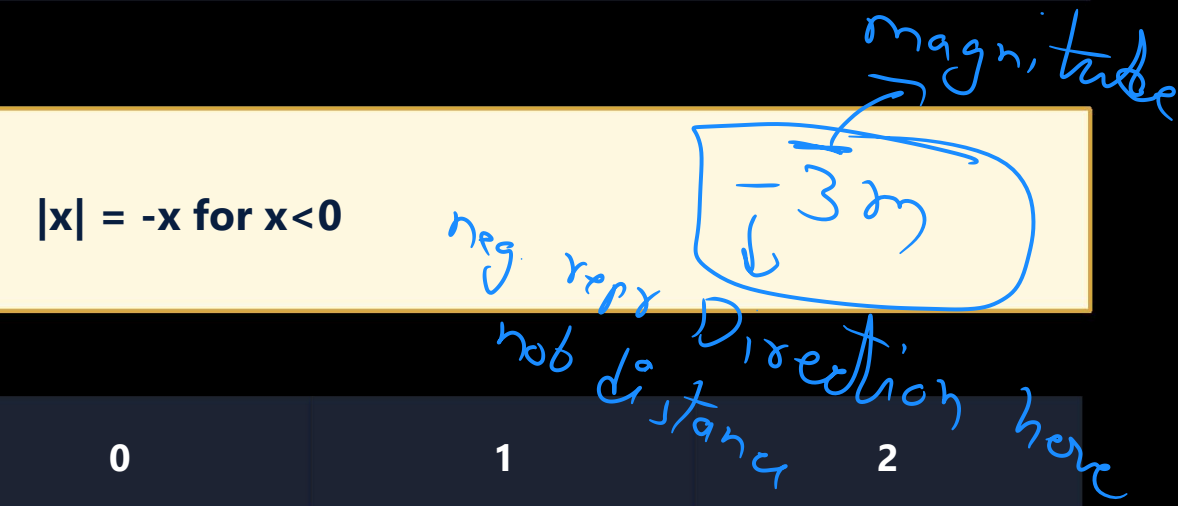
$$m_2 = \frac{-1}{3}$$

Meaning of modulus

Modulus gives distance from zero, so it can never be negative.

$|x| = x$ for $x \geq 0$ and $|x| = -x$ for $x < 0$

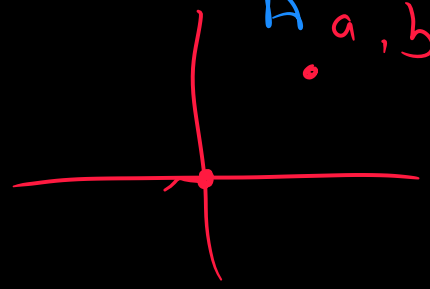
x	-2	-1	0	1	2
|x|	2	1	0	1	2



$|1| = 1$

$|-1| = 1$

$|x| = 3 \rightarrow x = 3, -3$



$|A| \Rightarrow$ Distance of point A from origin

Theory of modulus graphs

Modulus changes every negative output into its positive magnitude.

For x>=0: |x|=x, so use the line y=x.

For x<0: |x|=-x, so use the line y=-x.

The two straight parts meet where the expression inside the modulus becomes zero.

For y=a|x-h|+k: vertex=(h,k); sign of a gives up/down and |a| gives steepness.

$|-3| = 3$

$|x| = -1 \times x$

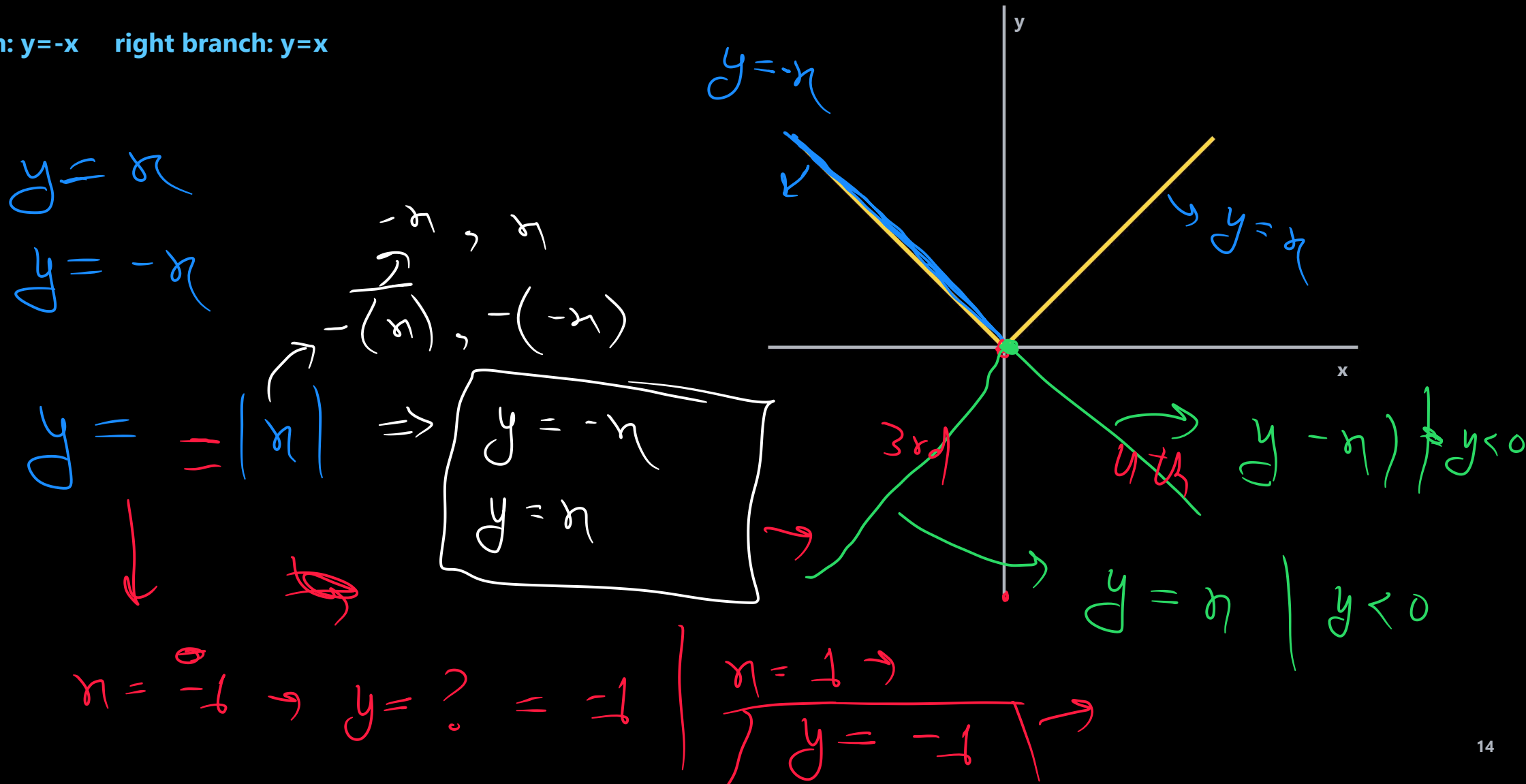
x is negative

by multiply x with -1 we can convert it into positive value

Graph of y = |x|

The graph is V-shaped, has vertex (0,0), and lies on or above the x-axis.

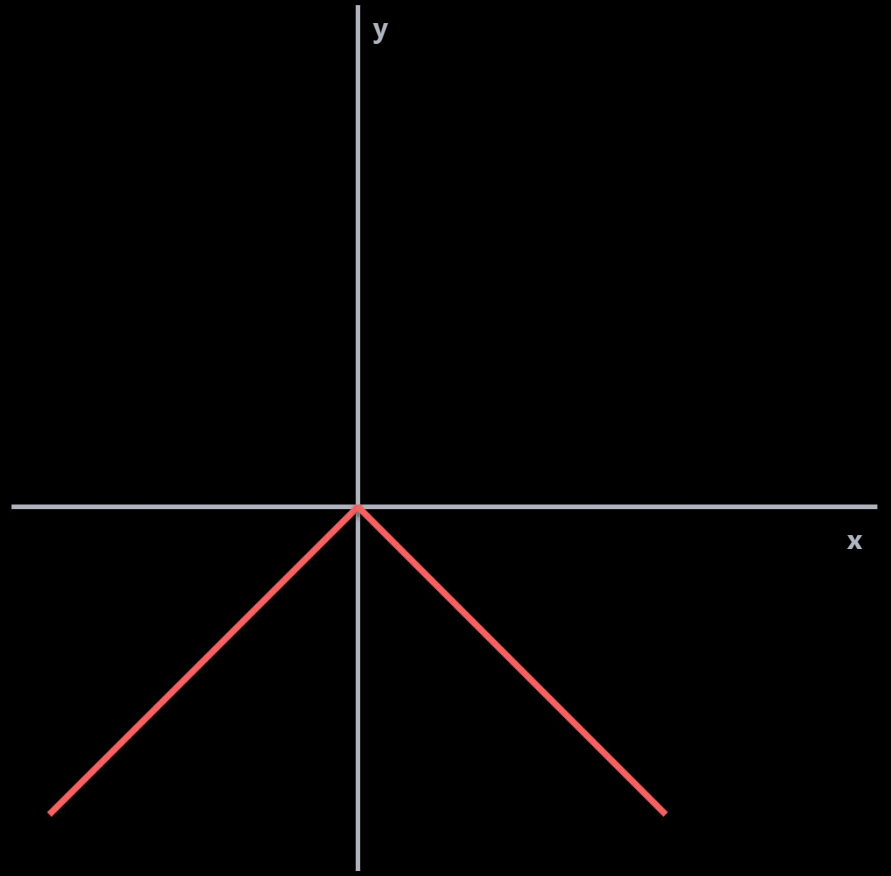
left branch: y=-x **right branch: y=x**



Graph of y = -|x|

The minus sign reflects y=|x| in the x-axis.

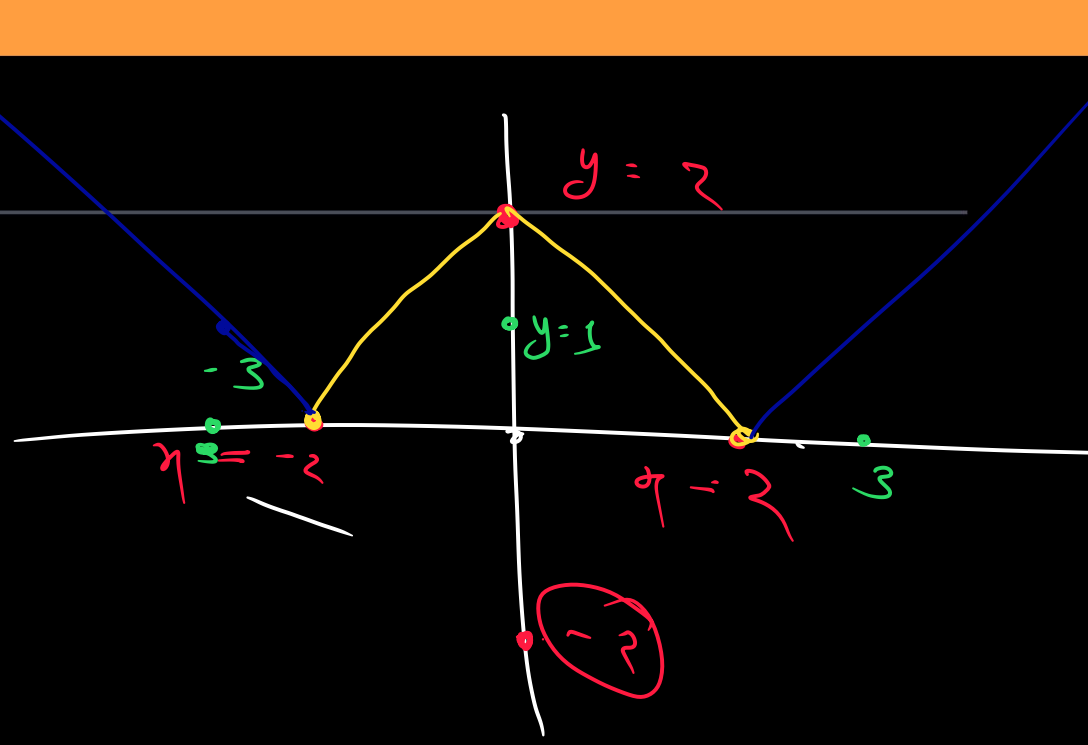
vertex stays at (0,0); the V opens downward



Draw y = |x-2|

$x = 3$

$y = 1$



Make a value table and draw y=|x-2|.

$$y = |x-2| \rightarrow \begin{matrix} (x-2) \\ -(x-2) \end{matrix}$$

At $x = 0 \Rightarrow y = 2$

At $y = 0 \Rightarrow x - 2 = 0 \Rightarrow x = 2$

& $-(x-2) = 0 \Rightarrow x = -2$

At $x = 3 \rightarrow y = 1$

$x = -3 \Rightarrow y = 1$

Simple solution

x	0	1	2	3	4
y	2	1	0	1	2

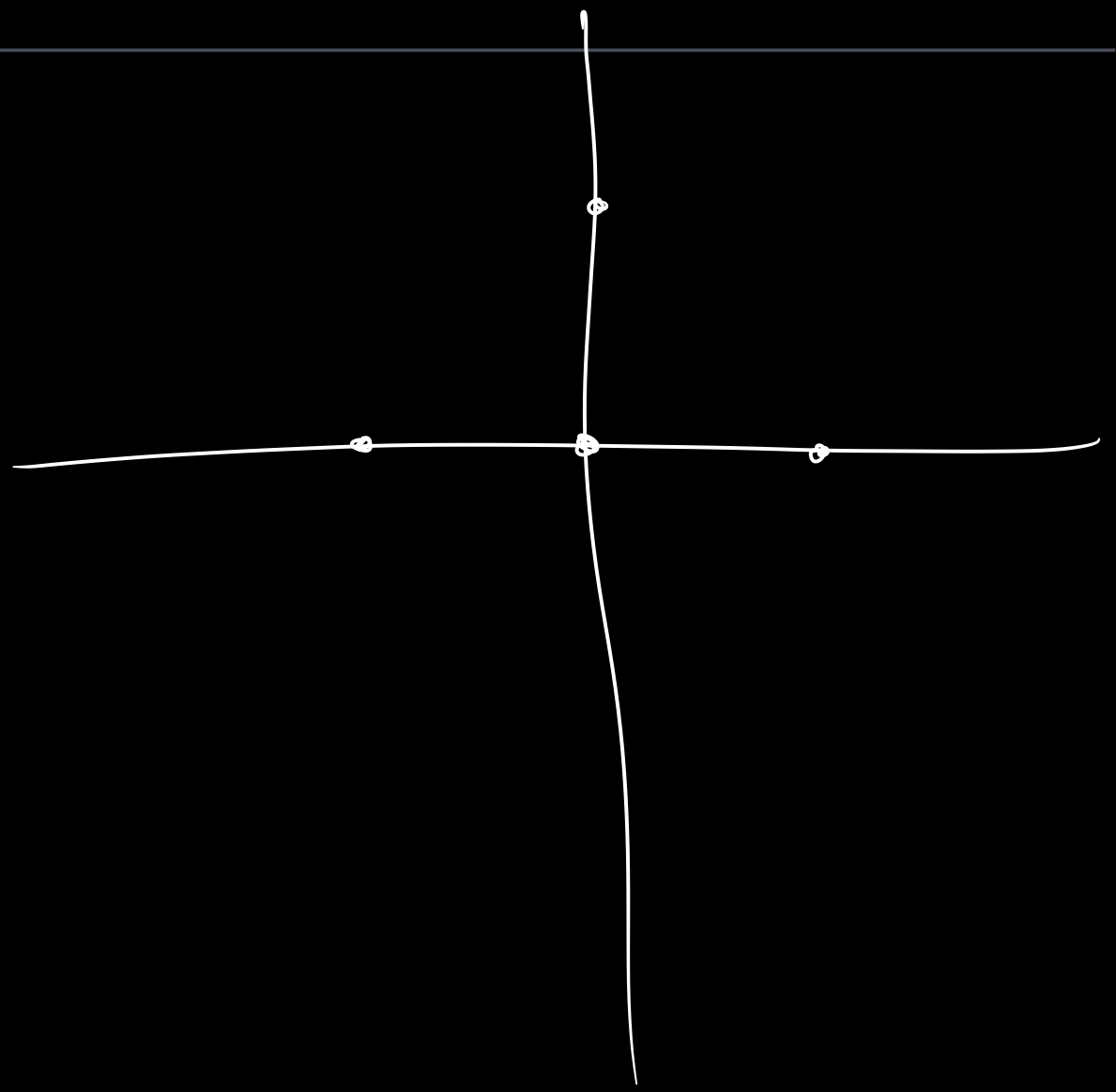
Step 1: The smallest value is 0 at x=2.

It is y=|x| shifted 2 units right; vertex = (2,0).

Rough work / extra work

$y = |x - 3|$

$y = |x + 3|$



Draw y = |4-2x|



Find the vertex and draw y=|4-2x|.

Simple solution

Step 1: Set the inside equal to zero: $4-2x=0$

Step 2: $x=2$, so the vertex is $(2,0)$

Step 3: $y=|4-2x|=2|x-2|$

V-shape, vertex $(2,0)$, steeper than $y=|x-2|$.

Position and steepness of modulus graphs

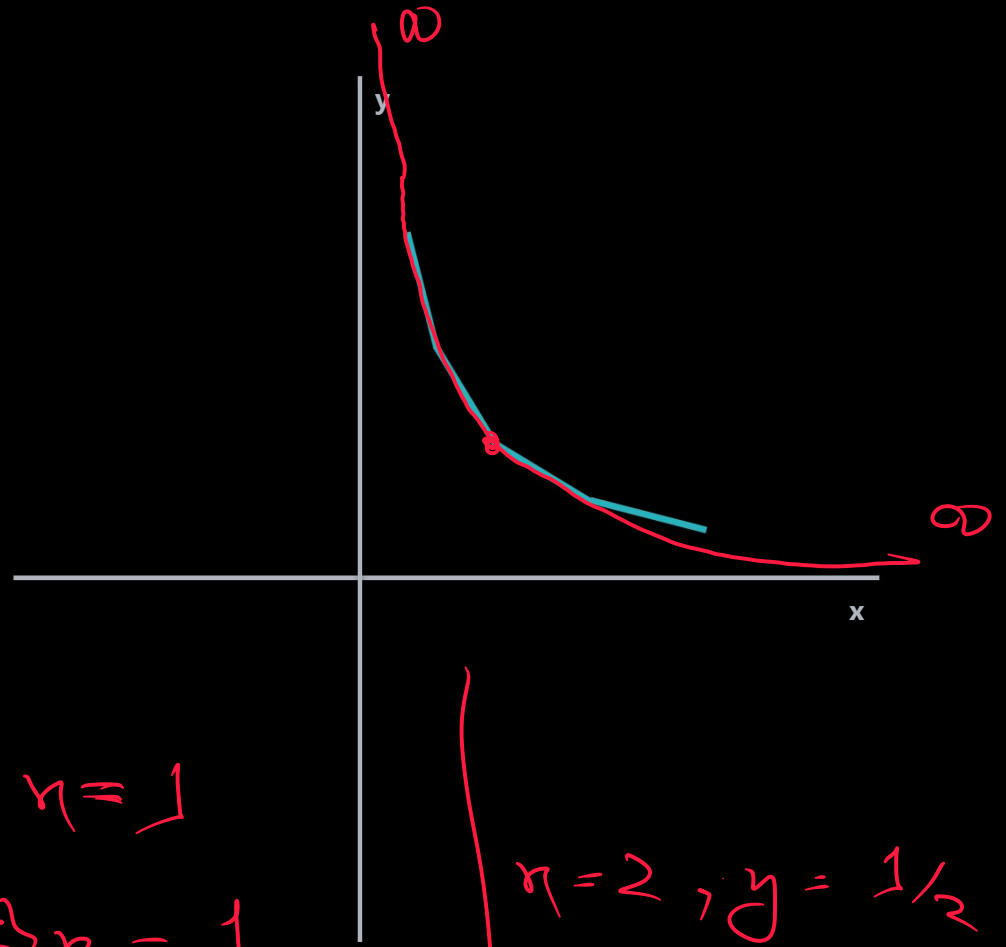
equation	vertex	opens	steepness
y=|x|	(0,0)	up	normal
y=-|x|	(0,0)	down	normal
y=|x-2|	(2,0)	up	normal
y=2|x-2|	(2,0)	up	steeper

Rectangular hyperbola: y = k/x

In an inverse relation, one quantity increases while the other decreases.

y = k/x or xy = k

The curve approaches both axes but never touches them.



$y = \frac{1}{x}$

$xy = 1$

$y = 1, x = 1$

$y = 2 \Rightarrow x = \frac{1}{2}$

$y = 5 \Rightarrow x = \frac{1}{5}$

$x = 2, y = 1/2$

$x = 5, y = 1/5$

How to draw an inverse graph

1. Write the equation in the form $y=k/x$.

2. Make a value table. Avoid $x=0$ because division by zero is undefined.

3. Mark positive and negative x values.

4. Join smoothly; do not let the curve touch an axis.

Rough work / extra work

Inverse-square relation

A force follows F=k/r^2. What happens when r becomes 2r?

$F = \frac{k}{r^2}$ ⇒ what happens when $r_{new} = 2r$

$F_1 = \frac{k}{4r^2}$ → F

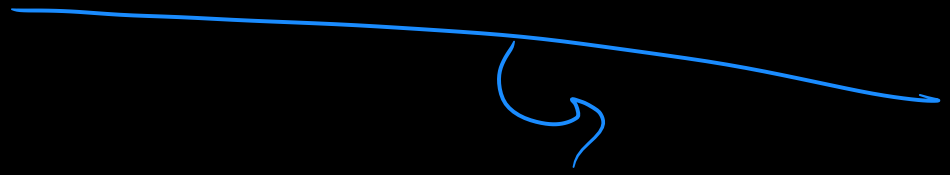
$F_1 = \frac{F}{4}$ ⇒ reduce F by 4 times

Simple solution

Step 1: Original force: F=k/r^2

Step 2: New force: F'=k/(2r)^2=k/(4r^2)

F'=F/4. Doubling distance makes force one-fourth.



De Broglie wavelength

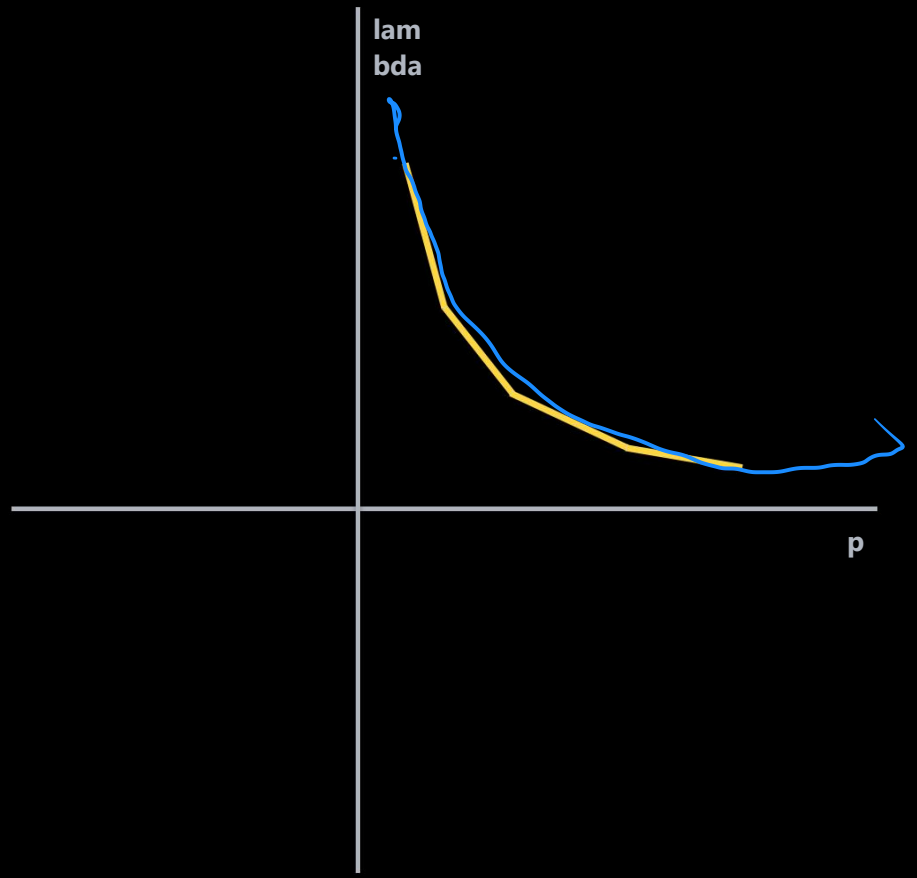
chapter - 13 / class - 12

A moving particle has wavelength inversely proportional to momentum

lambda = h/p

So lambda versus p is a decreasing rectangular hyperbola.

$$\lambda = \frac{h}{p}$$



Wavelength and kinetic energy

K=p^2/(2m) gives p=sqrt(2mK)

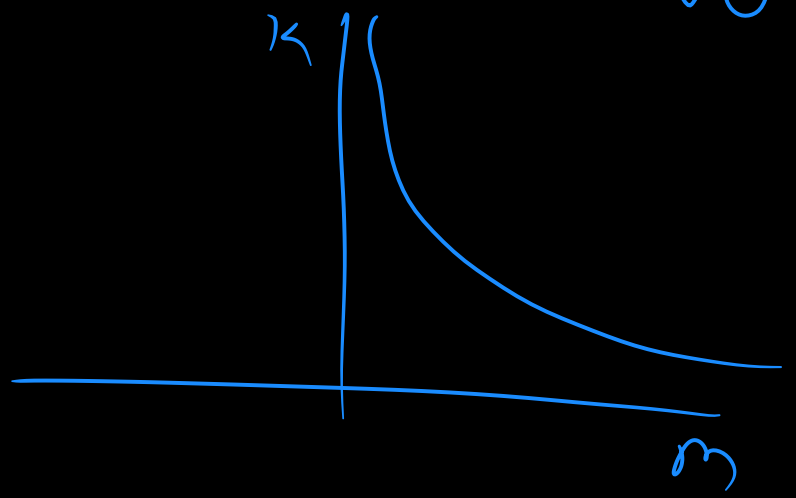
lambda=h/p=h/sqrt(2mK)

lambda is proportional to 1/sqrt(K)

Therefore lambda decreases when kinetic energy increases.

$$\frac{K}{\lambda} = \frac{p^2}{2m}$$

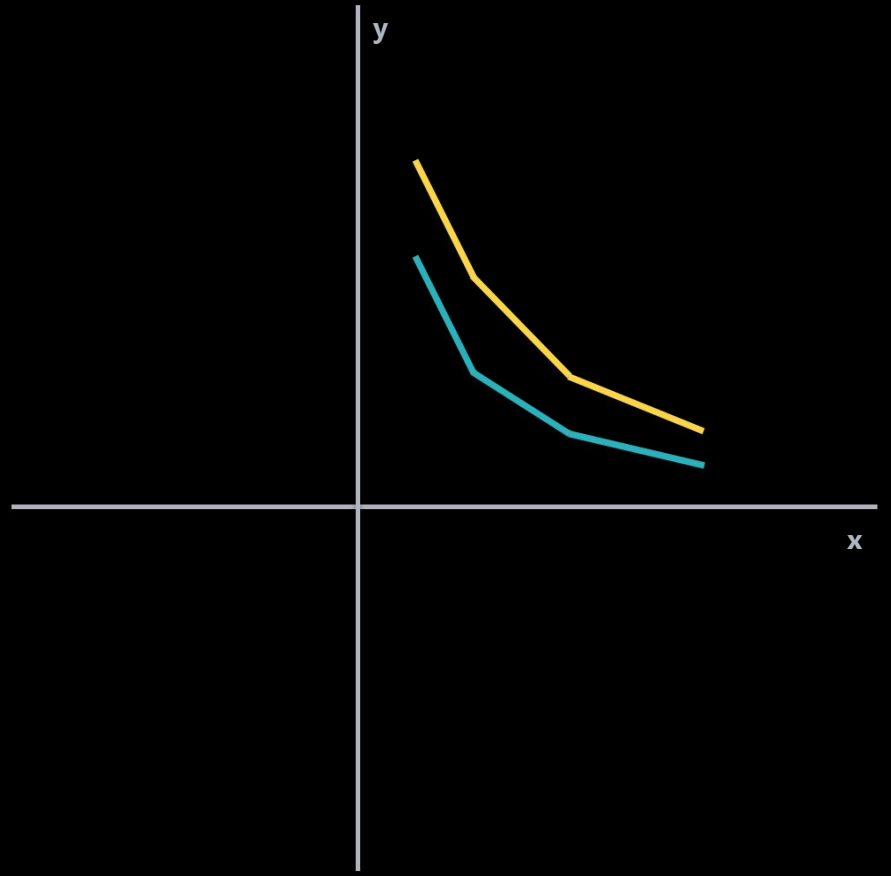
$\Rightarrow$ Kintic Energy



y = k/x for different k

Every positive k gives a similar decreasing curve.

A larger k places the curve farther from the origin.



Rough work / extra work

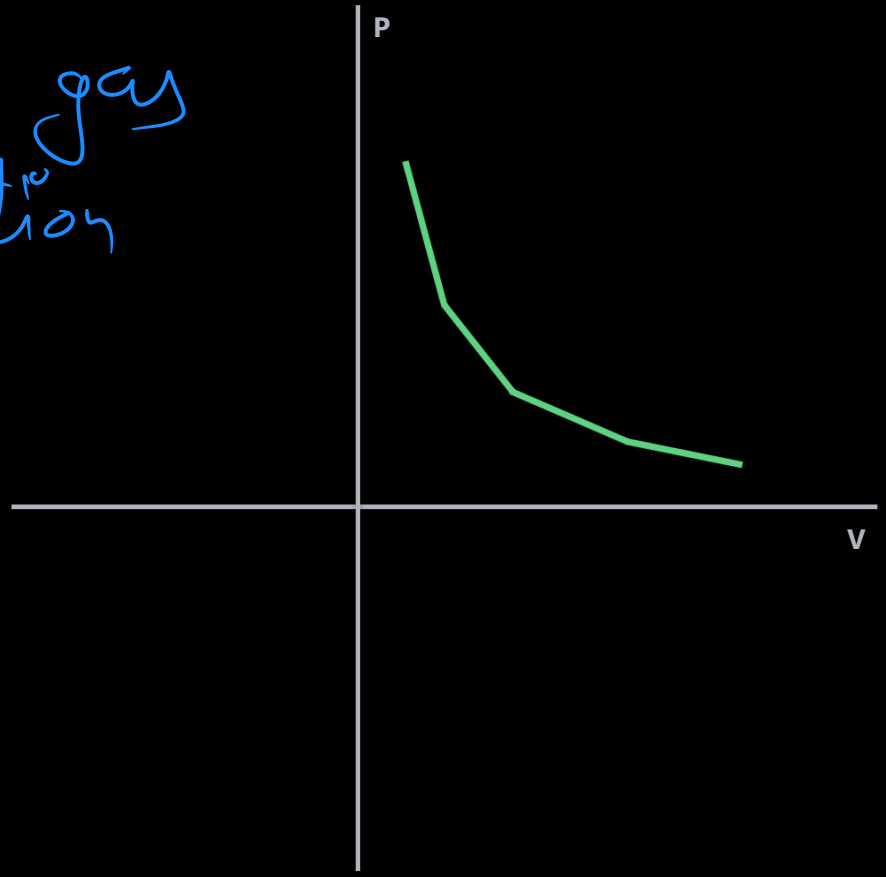
Pressure-volume isotherm

no need learn This

For a fixed amount of ideal gas at constant temperature:

PV=nRT gives P=nRT/V → Ideal gas Equation

P is inversely proportional to V, so the graph is a rectangular hyperbola.



Theory of an isothermal P-V graph

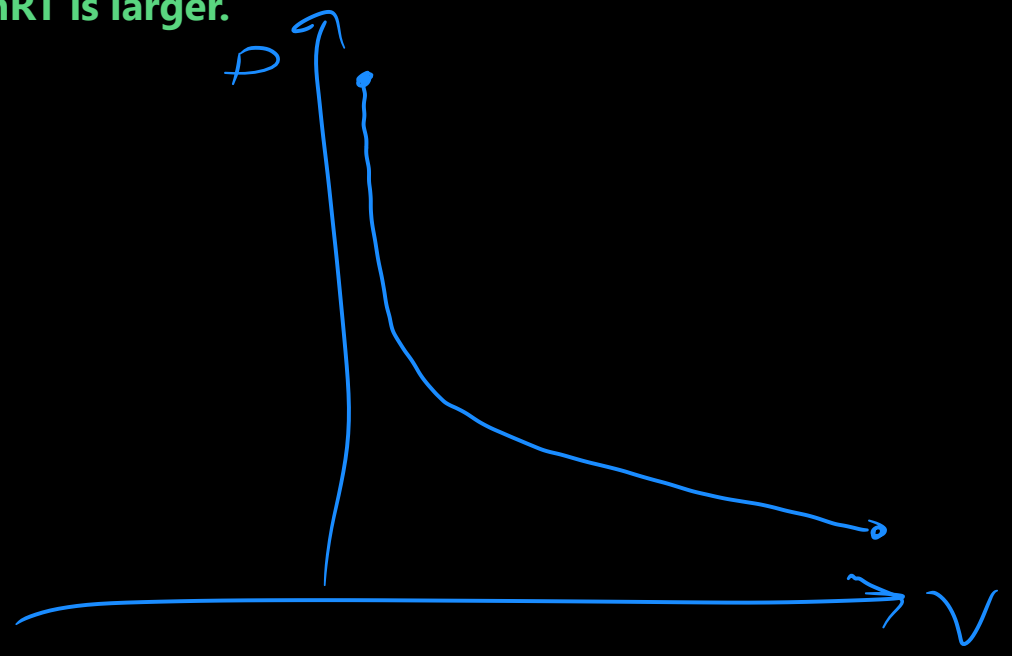
The amount of gas n and temperature T stay constant, so nRT is one constant.

PV=constant therefore P=constant/V

If volume doubles, pressure becomes half. If volume triples, pressure becomes one-third.

Pressure becomes very large near V=0 and approaches zero for very large V.

A higher-temperature isotherm lies farther from the origin because nRT is larger.



Two gas isotherms

Two P-V curves are drawn for temperatures T1 and T2. The T2 curve is farther from the origin. Which temperature is higher?

Simple solution

Step 1: At the same V: P=nRT/V

Step 2: Higher T gives higher P.

The farther curve has the higher temperature, so T2>T1.

y=1/x and y=1/x^2

feature	y=1/x	y=1/x^2
sign	same as x	always positive
branches	quadrants I and III	quadrants I and II
at x=0	undefined	undefined
far from origin	approaches 0	approaches 0 faster

$y = \frac{1}{0} = \infty$, $y = \frac{1}{0.1} = \frac{1}{0.2} = \infty$

Rough work / extra work

Electric field versus distance

Class 12

source	field relation	graph type
infinite line charge	E proportional to 1/r	rectangular hyperbola
point charge	E proportional to 1/r^2	inverse-square curve
electric dipole far away	E proportional to 1/r^3	steeper inverse curve

Larger power of r in the denominator means faster decrease.

for line charge $= \frac{1}{r}$

for point charge $= \frac{1}{r^2}$

Theory of electric-field falloff

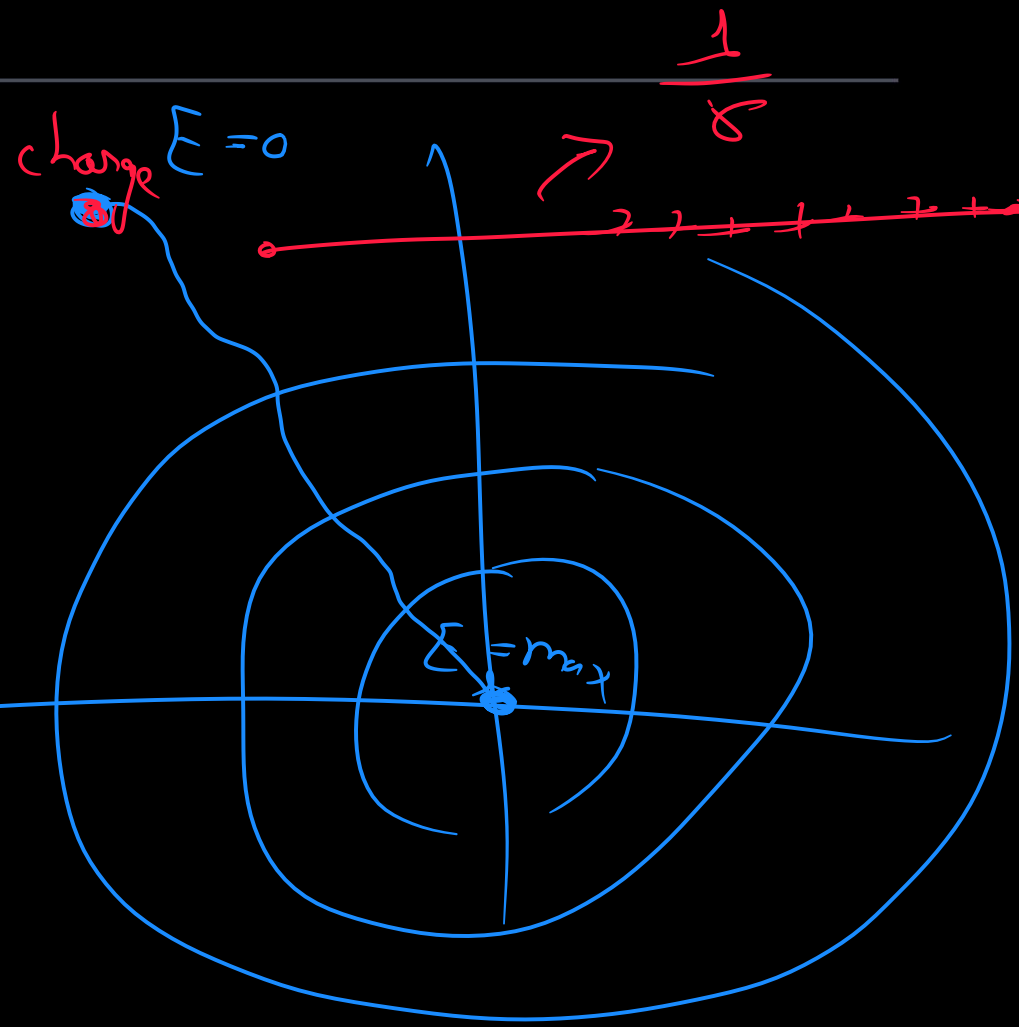
As distance increases, the same electric effect spreads over a larger region.

line charge: E is proportional to 1/r

point charge: E is proportional to 1/r^2

dipole far away: E is proportional to 1/r^3

At large r, 1/r^3 decreases fastest and 1/r decreases slowest.



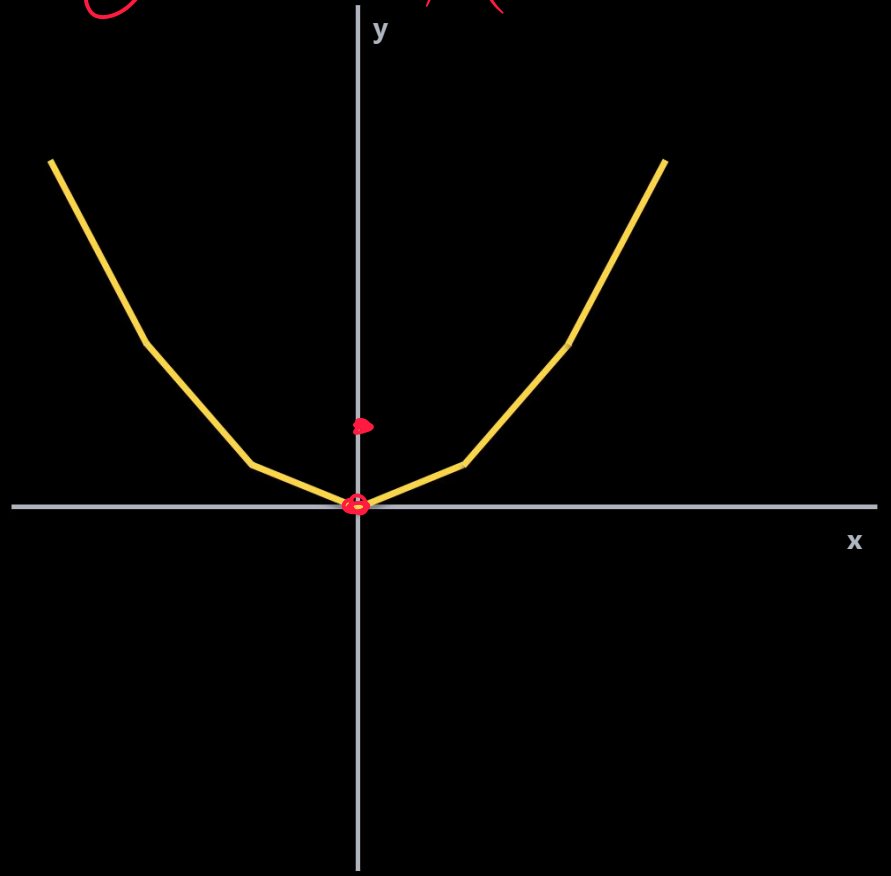
Parabola: y = x^2

$y = |x| \rightarrow$ V- shape

$y = x^2 \rightarrow$ U shape

Squaring gives the same y-value for +x and -x.

x	-2	-1	0	1	2
y=x^2	4	1	0	1	4



$y = x^2$

$x = 1, y = 1$

$x = 2, y = 4$

$x = -2, y = 4$

Theory of a parabola

The general vertical parabola is y=a(x-h)^2+k.

vertex=(h,k) **axis of symmetry: x=h**

a>0: opens upward **a<0: opens downward**

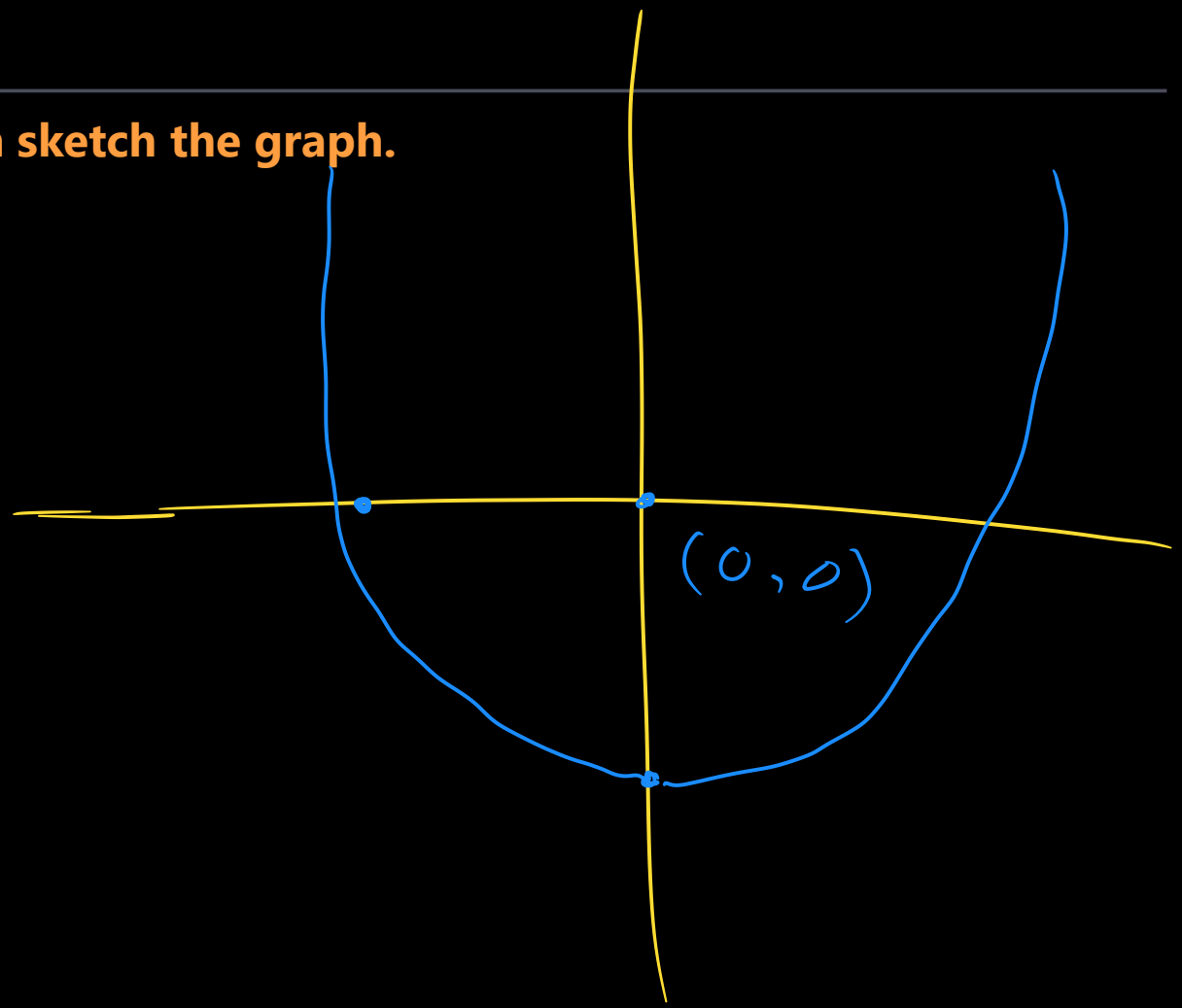
A larger |a| makes the parabola narrower; a smaller |a| makes it wider.

Find the vertex first, then calculate symmetric points on both sides.

Draw y = x^2 - 4

Find the vertex and x-intercepts of y=x^2-4, then sketch the graph.

$$y = x^2 - 4$$



Simple solution

Step 1: Vertex: x=0 gives y=-4, so vertex=(0,-4)

Step 2: For x-intercepts put y=0: x^2-4=0

Step 3: x=+2 or x=-2

The parabola opens upward and cuts x-axis at (-2,0) and (2,0).

Rough work / extra work

Vertical shifts of y=x^2

equation	vertex	direction
y=x^2	(0,0)	basic
y=x^2+4	(0,4)	4 units up
y=x^2-4	(0,-4)	4 units down

Downward parabolas

A minus sign before x^2 reflects the graph in the x-axis.

equation	vertex	opens
y=-x^2	(0,0)	down
y=-x^2+4	(0,4)	down
y=-x^2-4	(0,-4)	down

Practice

Draw every graph once using a small value table.

Label both axes and mark important points or intercepts.

Complete the circle and S.H.M. questions without looking at the answer slide.

Rough work / extra work